

VAC Forming

1. Two types of glue corresponding to two types of plastic i.e, ADS and Polypropylene.
2. Three rollers: two steel rollers (applies adhesive) and one rubber roller (compression roller).
3. Glue weight depends on Vinyl thickness, roller compression etc.
4. Glue Weight Quality Check: Color, Lot Number (left, center and right) and Amount of glue is measured. This is done every time roll is changed.
5. Glue weight ranges from: 0.8-1.1gms
Vinyl Thickness: 2.8-3.2mm

Roll Coating Section

1. Glue weight adjustment: Height adjustment can be done for qty of the glue.
2. Has a sensor that regulate qty of glue.
3. This section is in a roll coating room with Temp and Humidity that are critical to monitor.
4. There are camera sensor that points out whenever there are glue lines due to downtime, the sensor then instructs to cut that part.

Grippers and Clamp Frame

1. Grippers that get the vinyl sheets
2. gives a clean cut
3. Pulls up the table to the clamp frame
4. Clamp frame grips the vinyl
5. And proceeds to oven for heating

Oven

1. IR heating Bulbs
2. Both top & Bottom is heated
3. It is critical to monitor this temp
4. If we heat too much, then defects can occur.
5. heating is dependent on black and other colors, for black it is a bit cooler.
6. Heat profile is a bit different for each part type and mold combination

A very small cooling time is here

Vacuum presser

1. Clamps are gradually released as vinyl elongates on heating. So gradual releasing ensures flat vinyl on the tool avoiding any wrinkles and so.
2. For avoiding front & rear wrinkles stretch should be lengthwise. For avoiding wrinkles down the side, stretch should be cross wise.

1. Have three hoses that pumps cool air on the top side and a vacuum at the bottom. This leads to a pressure based pressing of the vinyl on the part.
2. each tool has two vacuum cups that holds the part. The vacuum % profile for each of them must be identical.

Types of defects:

1. Wrinkles (can be tackled by stretching at clamps)
2. Surface area defects (Bumps): dont know the reason
3. Glue lines: usually from downtime or unusual stoppage

Downstream Process

For stocking one part out of the four, all the four parts needs to be run.
cost: \$4 per pc of plastic part: $4 \times 4 = \$16$ for each part
\$13 for per piece vinyl: \$52 for each part

Total: $52 + 16 = \$67$